OBJECTIVE

This article describes standard guidelines for use by all I&FS District Offices to customize their own Pre-Installation Meeting Guides.

INTRODUCTION

Pre-Installation Meetings are held to bring together those people responsible for the successful installation of a General Electric Gas/Steam Turbine unit.

The meeting objectives are as follows.

1. To discuss the information described in this installation plan.

2. To establish a mutual understanding of commitments.

3. To describe General Electric's installation support activities.

4. To establish a channel of communication during the installation period.

To accomplish these objectives, topics ranging from a point in time prior to shipment through start-up will be covered.

Active participation and free exchange of ideas are encouraged at these meetings. GE representatives from both Power Systems and Energy Services will be present and qualified to answer questions and discuss matters of importance.

To maintain a common source of information available for future reference on the subjects discussed during this meeting, it is suggested that the minutes of this meeting be recorded and inserted at the end of this manual. Comments are welcome at any time.

AGENDA

I. Installation Support

II. Material Control

III. Installation Control

IV. Additional Services Available

V. Closing Remarks

VI. Appendix A

TO THE CUSTOMERS OF GENERAL ELECTRIC

This Pre-Installation Planning document is furnished by General Electric as a guide for installation and start-up of a Gas or Steam Turbine unit. It is not a complete instruction, but a compilation of the major factors to be considered by Purchasers and their agents in planning for an installation.

Installation and start-up of this type equipment is complex and involves numerous variables related to the interface of the gas or steam turbine with the Balance of Plant (BOP) installation. It is the responsibility of each Purchaser to develop a specific installation and start-up plan for their unit. General Electric Field Engineers will be available to assist in this planning.

The information presented herein is furnished without charge as an aid for installation planning. In no case should it be considered contractually binding or a complete list of all considerations.

The ultimate responsibility for installation and start-up of the turbine rests with the Purchaser. General Electric's responsibility is limited to providing technical direction as defined under the terms of the Equipment Contract.

I. INSTALLATION SUPPORT

An efficient, high quality installation requires Management, Labor, Supervision, and Technical Direction to supplement the materials purchased. Installation management, labor and supervision are the Purchaser's responsibility. Technical direction is provided by GE under the terms of the equipment sales contract. All labor, supervision, tools and equipment that are supplied under a separate installation contract should be coordinated with the technical data and guidance provided by GE under technical direction.

The following description of Technical Direction of Installation (TDI) is an interpretation of the equipment sales contract definition. It is General Electric's intention to provide Technical Direction of Installation using this interpretation as a guideline.

A. Technical Direction of Installation (TDI) – Definitions

Technical Direction of Installation for Gas and Steam Turbine equipment is an active, on-site service by General Electric providing engineering and technical guidance, advice, and counsel to the Purchaser and agents, with the objectives of providing a technically correct, high quality, safe installation. Work that is supervised, managed and performed by others is audited, observed, evaluated, and reported, with comparisons being made to applicable specifications, instructions, drawings, recommendations, current practices, procedures and techniques of GE.

It is the Purchaser's responsibility to provide direct supervision, management, regulations, arbitration and measurement of the Purchaser's personnel and agents regarding the related work. The Purchaser is also responsible for planning and scheduling the work. Guidance, advice and counsel provided to the Purchaser's personnel and agents in these areas are considered to be aspects of technical direction regarding the proper and timely installation and start-up of the gas or steam turbine equipment.

The GE representative providing TDI will be the primary contact for questions, problems, etc., regarding installation. When technical direction has been suspended, the I&FS District Office in _____ will act as the primary contact.

1. Technical direction provides full straight time service for a total period of up to ___ manweeks for the gas turbine and ___ mandays for the steam turbine. Straight-time service is the first 8 hours of each shift, 5 days per week, Monday through Friday, excluding recognized holidays. During those periods when no productive installation work is being accomplished on the involved equipment, technical direction may be suspended. One or more field representatives, as deemed necessary by GE, will be assigned to provide the TDI function.

   It is GE's intent to provide sufficient technical direction to cover the installation and start-up activities, beginning with the setting of sole plates or other equipment shipped in advance of the turbine, and ending with the completion of the installation work and start-up activities.

2. If the installation schedule is delayed or extended, and the duration of up to ___ manweeks for the gas turbine or ___ mandays for the steam turbine is exceeded, GE will render additional billing at GE's then-current Published Rates. When there is no appreciable activity for periods in excess of 40 work hours, the field representative(s) will be assigned to other jobs. In general, delays of this type most likely occur:

   a. Between arrival of equipment shipped early and preparation for setting of the sole plates.

   b. Between completion of the gas/steam turbine assembly work and plant start-up.

   Field service consultation will be made upon the Purchaser's request, and a field representative(s) will be returned to the jobsite upon reasonable notice.

3. Travel and living expenses, brought about by reason of work extension and/or suspension, will be billed at a published per diem rate.

4. If the installation schedule is accelerated and the overall job duration is shortened by multiple shift operations, there will be no additional cost to the Purchaser for the first 8 hours of each shift, 5 days per week, Monday through Friday. It is expected that the Purchaser will provide sufficient labor force on each shift to obtain optimum productivity within the above framework. GE will provide representatives to cover each shift.

   If the installation schedule is accelerated and the overall job duration is shortened by overtime operations, the Purchaser will be billed in accordance with the equipment sales contract, i.e., normally the premium portion of GE's then-current Published Rates.

   It is Turbine Contracts responsibility, in conjunction with Product Service Installation, to decide what action should be taken on behalf of I&FS in all matters relating to TDI. Contracts personnel will approach the customer as necessary and make all contractual adjustments to the original equipment order. I&FS field representatives will support contracts as requested, though recognizing that the TDI contract is between the customer and the Turbine Department.

B. Technical Direction of Installation (TDI) – Responsibilities

GE considers its TDI responsibilities to be those associated with making the gas or steam turbine unit ready for start-up and bringing the machine to full load. TDI participation in the installation and start-up activities varies by installation as required to meet the stated objectives of the project. The scope of this variation is largely determined by the requests, capabilities, and constraints of the Purchaser and agents.

The installation and start-up of a gas or steam turbine encompasses a network of related activities. Comments on consolidated groups of these activities are used here to further define GE's and the Purchaser's obligations for TDI, and to clarify its scope.

5. Pre-Installation Meeting – A pre-installation meeting is normally held before shipment of the major alignment components. The objective of this meeting is to ensure that the Purchaser and the Installation Contractor fully understand their responsibilities versus those of the Technical Director. Important items to review in detail at this meeting are the agreed upon mutual commitments, the Purchaser's installation schedule, the establishment of jobsite communication channels, the Technical Direction Staff, etc.

6. Unloading Material – TDI includes providing technical information and counsel for the unloading of all gas/steam turbine components at the location specified in the equipment sales contract, plus observation and auditing of critical phases of these

activities. Technical information includes dimensions, weights, lifting locations, special equipment, and GE recommendations.

7. Material Movement (i.e., skidding and cribbing of heavy components) – TDI includes providing technical information and counsel for material movements, plus observation and auditing of critical phases of the movements at the jobsite.

8. Examination of Material for Loss or Damage – The Purchaser or agent is responsible for examining all incoming material for loss or damage. TDI includes audit, verification, advice and counsel in all situations involving shortage, damage or loss, and assistance in filing claims with the carrier. Observation and audit of the unloading of major components includes attention to the condition of the material.

9. Inspection of Purchaser's Storage and Warehousing Facilities – The objective is to ensure an adequate receiving, storage and dispensing system to avoid damage to the stored material and compromising the installation plan due to losses or difficult retrieval. Off-site or long term storage entails special circumstances and is not normally included as part of TDI responsibilities.

10. Assistance in Maintaining Adequate Storage Records – TDI includes recommendations on record keeping, explanations of GE documentation and number systems, and assistance in identifying material.

11. Coordination and Administration of the Procurement of Tools and Equipment – Guidance in this area is provided to avoid compromising the installation plan because of items being unavailable at the proper time.

12. Checking of Foundation and Setting of Foundation Plates – TDI covers auditing of elevations, foundation dimensions, bolt locations, provisions for piping and wiring, potential interferences, and other similar details, plus technical guidance for and observation of grouting of foundation plates.

13. Installation, Inspection and Assembly of the Gas/Steam Turbine and Associated Equipment – This activity, consisting of several hundred actions, comprises the bulk of the installation process. Timely auditing and observation are required in order to provide TDI guidance, advice and counsel. Availability and utilization of technical recommendations, instructions and specifications are confirmed. Deficiencies are noted and corrections recommended. Engineering checks and data specified by GE are obtained and evaluated. Procedures are observed and audited, and modifications negotiated when necessary.

The Purchaser has an obligation to consult the Technical Director in advance with respect to scheduling and performance of all installation work to furnish the Technical Director adequate opportunity to monitor the work in progress during regular working hours. GE's recommendations may sometimes present installation difficulties. Where possible, the Technical Director will resolve such difficulties. When alternate solutions are not acceptable, the Technical Director will advise the Purchaser, in writing if necessary, of the risks involved should the Purchaser elect to proceed contrary to GE's recommendation.

14. Reviewing and Reporting of Schedules and Sequence of Installation – The responsibility for preparation, review, and updating of the installation schedule lies with the Purchaser or agents. The Technical Director will make recommendations based on available information, and periodically report on the progress of the gas/steam turbine installation as compared to the original mutually agreed upon installation schedule. Acceptance of this advisory service varies by installation. Therefore, GE will prepare a report to ensure that the Purchaser is continuously aware of GE's evaluation of installation progress.

15. Inspection of Assembly, Clearances, Alignment and Cleanliness to the Extent Possible without Dismantling – The Technical Director will observe and audit activities which are the responsibility of the installation contractor whenever necessary to ensure that results meet GE's specifications and recommendations. Exceptions will be noted (in writing if necessary) and reported to the Purchaser for resolution.

16. Installation and Placement into Service of All Supplier Equipment – Guidance, advice, and counsel is provided to a supplier when considered desirable by the Technical Director to resolve a problem.

17. Advice, Counsel, and Auditing of Fluid and Auxiliary Systems Flushing and Cleaning Operations – The Purchaser or agent is responsible for installing and maintaining clean fluid and gas/steam systems, and giving final approval of cleanliness. The Technical Director will make recommendations based on procedures and flushing equipment employed and the condition of filters, screens, and samples as observed.

18. Specialized Technical Direction Services – As part of TDI, GE will provide Start-up Engineers to check and adjust control systems as required.

19. Completion of All Required Equipment Modifications – Technical Direction includes keeping records of and advising the need for completion of all equipment modifications recommended by GE after shipment. The Purchaser will be

specifically advised, in writing if necessary, of modifications which are mandatory before operating the equipment.

20. Advice, Counsel, and Auditing of Placing Equipment into Service – The Purchaser is responsible for start-up and operation of the gas/steam turbine and the schedule and procedures for achieving this. Technical Direction of start-up is active guidance, advice, and counsel to the Purchaser with the objective of providing a technically correct, safe, timely, and reliable start-up. A comprehensive GE Start-up Report and Checklist serves as the gas/steam turbine pre-operational and start-up document, data record, and written report. The objective is to ensure that the necessary adjustments, tests, measurements, etc. are made, recorded and reported, and that the gas/steam turbine is in good operating condition with all features and devices in working order and ready for service.

21. Field Tests and Analysis – TDI services for new unit installations do not include testing after initial unit start-up and synchronization. Such tests include, but are not limited to, field reliability testing, field performance testing using station instrumentation or modified ASME, site sound surveys, emission tests, ambient surveys, or dynamic simulation studies.

    If requested, I&FS will assist the Product Department in performing field testing or analysis work under the direction of others, and will receive a charge number from the Product Department for material and services rendered on a T&M basis.

22. Safety – The Purchaser and agents will be advised, in writing if necessary, of observed safety hazards, incorrect applications, inoperative conditions, malfunctions of safety features or devices, and hazardous operating procedures.

C. Purchaser's Scope of Responsibility

As previously stated, installation management, labor and supervision are the Purchaser's responsibility. The following general comments further define the Purchaser's obligations and scope of responsibility.

1. Material Arrival – For proper inventory control and prompt action on transportation claims, GE must be notified when materials arrive at the jobsite. Inspection and storage of materials are the Purchaser's responsibility.

   If the foundations are not ready upon equipment and material delivery, or if suitable storage facilities are not available at the jobsite, the Purchaser will provide alternate storage space, unload and place the equipment and materials in this storage space, protect it during storage, and transport it to the installation site as required.

2. Management and Labor – The Purchaser is to furnish fully qualified labor, including necessary foremen and superintendents for supervision.

3. Tools – All tools, equipment, facilities and devices required for the safe handling, storage and installation of equipment are the responsibility of the Purchaser or agent. For Special Tools, see Reference III, Item E.

4. The Purchaser is to provide the following GE Technical Direction Office Facilities.

   A secure, furnished, private jobsite office facility of sufficient size (near the equipment) to accommodate 3 to 4 Field Engineers, where drawings, special tools, and other GE Proprietary material and equipment can be safely stored for reference.

   The GE site office is to include at least two (2) direct phone lines, desks and chairs, print table, print rack, filing cabinet, bookcases, heat and air conditioning, potable water, adequate lighting, electrical outlets, and easy access to sanitary facilities. Adequate parking must be available for the Field Engineers.

   Note: GE Technical Direction office facilities should be agreed upon during the Pre-Installation Meeting.

5. Operating Personnel and Support – Operating personnel, compressed air, gas/steam, fuel, cooling water, Halon, electric power, oil, and supplies, etc. for starting, operating, and testing equipment are the responsibility of the Purchaser or agent.

6. Installation Schedule – The installation schedule should be complete at the time of the Pre-Installation Meeting. The responsibility for preparation, review and updating the installation schedule lies with the Purchaser. The Purchaser is advised to consult the field representative in advance with respect to scheduling and performance of all installation work to allow the field representative adequate opportunity to review the work in progress during regular working hours.

II. MATERIAL CONTROL

It is important to set up proper facilities and procedures for receiving, inventorying, and storing material at the jobsite. Following are recommendations for handling material upon receipt. Installation efficiency is maximized when material is readily available at the required time. While identification and accessibility are major factors, the need for quickly replacing damaged or missing material is equally significant in maintaining a sensible work plan. The methods outlined in this section are recommended to facilitate prompt replacements or repairs when required. It is standard GE practice to place priority on replacement orders that endanger start-up schedules. The Purchaser's cooperation in this area will help avoid installation schedule problems.

A. Arrival and Inspection

The Purchaser or agent is to inspect all material and equipment upon arrival at the carrier's final delivery point. The inspection is to include but not be limited to the following.

1. Inspect the protective covering and packing for obvious damage during transit.

2. Inspect the "Hump Meters" and blocking of components on the transport vehicle for indications of movement during shipment.

3. Check the orientation of any components relative to gas/steam turbine foundation where special orientation had been requested.

4. Inventory and check the "As Received" condition of all equipment and material.

5. The carrier's documentation is to be reviewed for the following before accepting delivery.

   - Material Safety Data Sheets (MSDS) for HAZARDOUS material, and a packing list for the equipment being delivered.

Carload shipments are to be promptly unloaded to avoid demurrage charges, and arranged in an accessible and convenient manner for further inspection and handling.

A detailed, systematic inventory control system is critical to the performance of an effective and efficient installation.

B. Material Inventory and Control

Following is the process for receiving MSD equipment and materials.

1. When notice is received from the supplier that equipment and materials are ready to ship, GE will issue to the supplier a Notification of Shipping Release (NSR) authorizing shipment to the job site. A copy of the NSR containing the information listed below will be faxed to the Customer's Material Representative.

   - Case Number
   - Package Type (Case, Crate, Skid, Box)
   - Weights and Dimensions
   - Vendor Name

   The most significant piece of information contained on the NSR is the list of Case Numbers included in the shipment. Each case is assigned a number using a systematic labeling method which specifically identifies the equipment and material being shipped.

   The first digit in the Case Number identifies the Unit with which the material is to be used. The next four alpha-numeric characters identify the Model List Item (MLI) number of the equipment. The last two digits are sequentially numbered (beginning with 01) to identify the particular case. For example, the first shipping case for the Inlet Duct Arrangement (MLI A041) for Unit No. 1 would be assigned the Case Number 1A04101. If 16 cases were needed to ship the entire arrangement, the remaining cases would be numbered 1A04102, 1A04103, etc. up to 1A04116.

   Using this detailed, systematic labeling method helps facilitate receiving and inventory processes, and can aid in planning site laydown areas.

2. The Bill of Lading (BOL) for each shipment received should list all the Case Numbers included in that shipment. Each case in the shipment should have a packing list on the outside and inside of the box. Upon receipt, the BOL should be checked against the cases received to ensure that all equipment and material intended for the shipment was actually delivered. Copies of the BOL and packing lists should be provided to the GE TA Staff.

3. A running tally of cases received vs. NSR's issued should be kept. Any missing or damaged items are to be reported to the GE TA Staff using procedures described in the following sections. A good practice is to use a site map to record and identify the locations of various cases. Generally, all the cases comprising a single item should be received within three weeks after issuance of the NSR.

C. Material Shortages

Shortages may appear as materials are received. "Shortages" are defined as any part which is needed but not available at the jobsite, and fall into one of four categories.

1. Material is known as shipped but is not received at the jobsite.

2. Material is not known as shipped but is known to be needed.

3. Material is received in insufficient quantity.

4. Material received is damaged in shipment, damaged on site, or lost on site.

D. Material Shortage Report

The Technical Director will issue a Material Shortage Report for the following cases.

1. An insufficient quantity of materials is received at the site.

2. Needed materials are not received and are not known to be shipped.

3. Materials are known to be shipped and are not received.

Note: The above cases assume that no fault lies with the carrier.

A Material Shortage Report, indicating the required information and reason(s) for the shortage, will order the required materials and initiate shipment to the jobsite. In these cases, GE will assume the cost of providing the needed shortage material.

E. Lost or Damaged Materials

When material is lost or damaged in transit or at the jobsite, the Technical Director must issue a District Requisition to order replacement material from the factory. Each District Requisition issued must reference either a customer Purchase Order or Notice of Claim against the carrier. Without this reference, the Requisition will be returned by Customer Service and the Requisition will not be processed.

When the Equipment Sales Contract calls for delivery of parts, F.O.B. (Point of Shipment), the Purchaser is responsible for "Risk of Loss" in transit. In such cases, the responsibility for filing and resolving transportation claims with the carrier lies with the Purchaser or agents. The Technical Director will require a Purchase Order in order to issue a District Requisition or an FSR for repairable damaged materials under the following circumstances.

1. Materials are damaged in shipment.

2. Materials are lost in shipment.

In those cases where contract terms state that GE retain the "Risk of Loss" for material in transit, GE will be responsible for filing and resolving transportation claims with the carrier. The Technical Director will issue a District Requisition for missing materials or an FSR for repairable damaged materials in the following cases.

1. Materials are damaged in shipment.

2. Materials are lost in shipment.

It remains the Purchaser's responsibility to notify GE when the above situations occur.

The Technical Director will require a Purchase Order in order to issue a District Requisition for missing materials, or an FSR for repairable damaged materials, under the following circumstances.

1. Material was received at the site and damaged thereafter.

2. Material was received at the site and lost thereafter.

Once materials are received at the site, they become the responsibility of the Purchaser or agent. The cost of replacing or repairing parts is the responsibility of the Purchaser or agent.

F. Transportation Claims

It is the responsibility of the Purchaser or agent to enter any and all claims against the carrier for any in-transit damage or loss of equipment, except where the Equipment Sales Contract specifically states the responsibility for "Risk of Loss" is to be assumed by GE.

The following reference information regarding transportation claims is provided for the agent responsible for receiving materials so that prompt action can be taken in the event that material is lost or damaged in transit.

A claim exists against the carrier for the full amount of loss or damage whenever material shipped by GE arrives in a damaged condition as a result of mishandling by the carrier, or when the quantity of material received is less than that shipped. This also holds true for "concealed loss or damage," in which an inspection of contents reveals damage or loss even though a box or crate is not broken and shows no evidence of mishandling.

Most loss or damage occurring in transit can be discovered at the time the shipment is received. Therefore, all freight is to be examined carefully before accepting delivery from the carrier. When loss or damage is found, make a suitable notation on the freight bill, ensuring that the carrier's copy contains the same information.

Notations should be brief but accurately describe the condition reported. Vague descriptions such as "carton open, carton rattling, carton crushed" are not sufficient.

1. Inspecting Merchandise – Visible Damage

    a. **Do Not** sign for material which has not been delivered.

    b. Check shipments against the freight bill to ensure that all merchandise listed on the freight bill is actually received.

    c. If possible, avoid notations such as "one short." It is better to identify the missing piece by describing the goods or listing the case, carton number, etc.

    d. If a carton rattles, or is torn or crushed, or shows other evidence of damage, it should be opened, the contents checked, and an inventory of the conditions found noted in detail on the delivery receipt.

2. Concealed Damage – Loss or damage discovered after delivery to the consignee shall be reported by the consignee to the carrier immediately upon discovery, and in any event within 15 days after receipt. The contents and container are to be held for inspection by the carrier with a statement of facts or circumstances evidencing loss prior to delivery by carrier. Inspection by the carrier is to be made when practicable, and in any event within 48 hours of notice, and shall include examination of package and contents for evidence of abstraction of the missing goods, checking contents with invoice and weighing for comparison with abstraction of the missing goods, checking contents with invoice and weighing for comparison with shipping weight, and an investigation of the cartman's record of handling shipment. Inspection Reports are to be made in duplicate on a standard form and signed by carrier's

agent and consignee, with one copy to be retained by consignee and attached to claim for loss, if made. If no inspection is made by carrier's agent, consignee's inspection shall be accepted as carrier's inspection.

3. While the Statute of Limitation is generally 9 months for filing a loss or damage claim against a carrier, it is important that a claim be filed as soon as all pertinent facts are available. Frequently it is a matter of weeks or months before repair costs become known.

   When loss or damage in transit is discovered, whether concealed or apparent, the carrier's agent must be notified immediately and a carrier inspection requested if a damage claim is involved. A copy of the carrier's inspection report should be obtained at the time of inspection. Such advance notice should be supplemented by the prompt filing with the carrier of a "Notice of Claim." The following additional papers will be needed when filing a claim.

   a. Original Bill of Lading (BOL).

   b. Original paid freight bill.

   c. Original invoice or certified copy.

   It is necessary that the carrier have these documents to establish that the claimant is the party entitled to make the claim, that the carrier to whom the claim is presented was involved in the movement, that it has been paid its freight charges, and that the amount claimed is not more than the actual value of the property shipped.

   When completing the Notice of Claim, provide a dollar estimate of the damage and where repairs will be made. Attach to the notice the carrier's inspection report, or a formal document indicating the carrier has waived inspection. A copy of the freight bill should be included if available.

4. United Parcel Service (UPS)

   a. United Parcel Service (UPS) treats all shipments on an F.O.B. destination basis. UPS will not accept claims from a consignee for loss or damage. Such claims must be made by the consigning point that placed the shipment with UPS for delivery.

b. Upon notification that a shipment has not been received or was received in a damaged condition, the consignee should immediately file a notice of loss or damage with the local UPS claim agent, and forward all details to the GE point of shipment. GE will then will file the claim for loss or damage with the UPS unit which originated the shipment.

c. When disposition of the loss or damage is agreed upon by the shipper and the UPS claim agent, the shipper may perform any of the following.

   1. Make a replacement shipment to the consignee.

   2. If practical, arrange to have material repaired and returned to consignee.

   3. Arrange for issuance of credit to customer if material damaged is not to be repaired or replaced.

d. Billing is to be issued by the field or factory to UPS for the amount due in any of these options. The invoice must contain the following information.

   1. UPS Shipper Number.

   2. UPS Claim Number.

   3. Name and title of UPS Agent giving final disposition.

   4. Date of original shipment.

   5. Brief statement as to loss or extent of damage. Also, attach to this invoice a copy of the original invoice.

III. INSTALLATION CONTROL

The size and complexity of gas/steam turbine equipment makes it unfeasible to fully assemble and test the equipment in the factory. To ensure that equipment is properly installed, aligned, and checked before operation, GE provides Technical Direction during installation. This service is essential for proper interpretation of the design criteria as expressed in drawings and instruction books. Equipment changes may be required in the field resulting from late requests from the Purchaser, mistakes, design improvements based on operating experience, etc. Additionally, GE is continuously evaluating installation methods. New or improved methods will be provided to the Purchaser and agents as they are developed. Formal transmittal of these changes from the Product Department to I&FS are through Field Modification Instructions (FMIs) or Technical Information Letters (TILs).

1. FMIs are limited to descriptions of equipment changes. An FMI will list the affected drawings and disposition of the affected equipment. Copies of changed drawings will accompany the FMI.

2. TILs are used when a generic change is involved, especially when a large distribution is required. TILs are also used to transmit installation recommendations in areas where drawing changes are not involved.

   GE uses several other methods of providing installation directions to the jobsite. Publications with identifying series numbers (such as GEI-86151, Low Voltage Power Circuit Breaker, and GEK-39672, Switchgear) are issued by cognizant engineering functions. Most of these publications are included in the Instruction Books distributed at the time the unit ships. There are some which are not directly related to specific equipment and thus are not included in the instruction book. These are distributed by the I&FS District Office when applicable to the installation.

A. Installation Schedule

A basic requirement for a successful installation is an installation schedule. The schedule can range from a bar chart to a complex logic network programmed with provisions for resource allocations. The bar chart allows for listing work activities in sequential order, planning job duration, determining labor and material requirements, and job tracking. The logic network when properly programmed allows all of the above, plus critical path planning, periodic job tracking with compensation for manpower and material deficiencies, job cost tracking, etc. The installation schedule used should be designed to meet the needs of the specific installation and to provide optimum installation performance.

The I&FS District Office will be available to assist and offer suggestions and comments on the installation schedule. However, the responsibility for preparation, review and updating the installation schedule lies with the Purchaser, the agents or the installation contractor.

1. Scheduling Craft Labor

   The installation contractor is responsible for scheduling craft labor.

2. Schedule Interfaces Between Contractors

   Certain interfaces between the gas/steam turbine and other equipment on-site must occur during the installation. Prior to interface activities and depending on the operations involved, a meeting should be held to discuss dates, problems, specifications, and any special requirements that may be involved. The Technical Director on-site will be available to attend and assist in the resolution of any

interference problems. Responsibility for scheduling and conducting such meetings lies with the Purchaser.

3. Progress Meetings

   Periodic reviews of the installation schedule are of major importance in achieving the commercial operating date. A Field Representative will provide full cooperation during all schedule review meetings, and will advise of the progress being made as related to the schedule. The Field Rep. will also be available to assist in updating the overall work plan to reflect any significant changes in the installation schedule.

B. Construction Prints

A Construction Print List (CPL) contains all the Primary and Secondary drawings, process instructions, applicable parts lists, and detailed machining drawings. These prints represent the "As Built" condition of the turbine and are provided for use by the Technical Director to assist in consulting functions. Drawings within the sets that are changed during installation will be replaced with updated drawings. When the installation is complete, the Technical Director will provide the Purchaser with a set of the construction prints without the parts lists (PLs) and machining drawings. This set will represent the "As Built" condition of the gas/steam turbine.

C. Extra Work Procedures

On occasion, there will be extra work effort required on the jobsite to correct or perform work which is normally a factory operation.

Additional Service Work is defined as work performed in the field during technical direction to correct results from any Product Department engineering or manufacturing error basic to fulfillment of the Equipment Sales Contract. It is classified as follows.

1. Completion to factory quality standards of any component shipped in an unfinished condition.

2. Field purchase of sales contract materials.

3. Local purchase of small hardware items to cover shortages.

D. Back Charges to General Electric for Additional Service Work

It is important that a common understanding exist of the parameters within which GE will pay direct costs for labor necessary to correct or complete work meeting the definition of Additional Work.

1. Labor – Except in those cases where specialized skills are required, GE will expect to employ the same craft labor used in the installation of the turbine-generator.

2. Assignment of Labor – The GE Technical Director only is authorized to request and assign labor to Additional Service work. The Technical Director on site will request the craft labor required to complete the additional work and issue a GE Purchase Order with an assigned identification number to the installation contractor.

3. Documentation Required for Payment of Back Charges – GE will not honor back charges unless properly documented as follows.

    a. A copy of the Purchase Order issued by the Technical Director must accompany the billing invoice.

    b. Copies of timesheets (signed by the Technical Director) verifying labor manhours expended to complete the particular additional work item associated with the Purchase Order must accompany the billing invoice.

    c. All documents reflecting the additional work must reference the original Purchase Order for that specific work.

    d. The I&FS District Office must receive the billing invoice no later than 30 days after completion of the work for which the Purchase Order was issued.

All parties involved with administration of craft labor assigned to the gas/steam turbine installation must be identified to I&FS District personnel prior to starting work on the unit. Details for administering labor costs and billing associated with additional work expense can then be established between the responsible parties.

4. Limitations of Labor Overtime for Additional Work Items

    Charges only for the straight time portion of hours worked by craft labor will be accepted by GE as labor costs associated with a Purchase Order for resolution of Additional Work. This policy also applies when the normal work schedule includes one or more hours of overtime per day.

    Craft labor should be scheduled to perform the work on these items during the normal straight time working hours of the Technical Director.

Only in rare circumstances where additional work activities will delay progress of the installation will craft labor overtime be authorized. This authorization must be obtained in writing from the General Electric Technical Director on a daily basis.

E. Special Tools

1. The installation contractor is responsibe for providing all normal tooling, rigging, equipment and devices required for safe handling and installation of the gas/steam turbine unit, and for providing facilities for equipment storage while it is being used on the installation. These tools include the test equipment necessary to install and calibrate the control system and associated devices, including but not limited to multimeters, oscilloscopes, signal generators and millivolt sources with associated leads and connectors.

2. GE will supply the special tools for gas/steam turbine and generator installation that are not normally furnished by the installation contractor.

3. A list of special tools provided by GE is included with the CPL under "Tool List" or "Installation Devices". The use, care and storage of this equipment while on the project site will be the responsibility of the installation contractor.

4. Special tools identified on these lists are part of the gas/steam turbine and generator package, and are to be turned over to the Purchaser (and documented) upon completion of the installation. It is important that these devices be carefully stored and protected for future maintenance work on this machine.

5. General Electric will furnish drawings and sketches for any special tools that are needed on a job-specific basis.

IV. ADDITIONAL SERVICES AVAILABLE FROM I&FS

General Electric's Installation and Field Services (I&FS) organization has a variety of field services available to the power generation industry. These services may be purchased to complement the Technical Direction of Installation provided under the Equipment Sales Contract, and to provide a broad base of "In-Service" and maintenance service programs. The following section describes services available I&FS.

A. Field Engineering Services

Field Engineering Service is engineering guidance and technical advice provided to the Purchaser regarding current engineering, manufacturing, installation and operating practices relating to the GE equipment involved in the installation.

B. Management Service

Installation/Maintenance Management Service can be any combination of the following services: planning, monitoring, scheduling, preparing management reports, and selecting and assigning craft labor. This service provides full-time coverage by field representatives on a continuous basis from the beginning of the GE equipment installation/maintenance work at the site until the equipment is turned over to the Purchaser for operation.

C. Management Service with Responsibility for Craft Labor

This service is the same basic work scope as described in Item B. above, but also includes responsibility for the conduct and quality of work performed by craft labor. It includes in its scope selecting and supervising the craft labor crews.

D. Complete Installation/Maintenance Service

This service is a combination of planning, management, labor tools, and miscellaneous materials to move, install, assemble, connect, interconnect, and start up and/or maintain GE equipment.

E. Engineering Study/Inspection/Test Service

This is a specialized service performed on equipment or systems in accordance with industry practices by experienced personnel using special instrumentation and devices where required with the objective of reporting expert opinions and recommendations related to the current condition and future service ability of the equipment or system.

F. Dedicated Field Engineering Service

This service provides full time coverage at the Purchaser's plant for services to be used by the Purchaser at the Purchaser's discretion.

G. Training Programs

In addition to operator training, GE also has optional training programs available. Examples of courses include the following.

- Operator Training
- Gas Turbine Maintenance
- Speedtronic Control Systems
- Technician Training
- Steam Turbine Maintenance

Arrangements should be made with the local I&FS office to tailor courses to specific customer needs.

Note: A discussion should be held during the Pre-Installation Meeting to define the training programs purchased by the customer as a part of the equipment contract and to schedule performance of the courses.

V. CLOSING REMARKS

The minutes of the Pre-Installation Meeting will be distributed to all attendees. It is suggested that the minutes be inserted under the EM 5005 tab for future reference. All action items or questions incumbent upon GE will be answered prior to the actual start of installation activities.

The Purchaser, consultants and contractors are encouraged to carefully consider the material presented in this manual. Some of the planning to be considered by the Purchaser or Purchaser's representatives will require action in the near future in order to be effective. The I&FS district representatives are available to answer questions and provide guidance. Referencing this document is encouraged when installation planning questions arise.

A. Future Orientation Sessions

In addition to participating in the Pre-Installation Meeting, additional jobsite meetings will be scheduled on an as needed basis. Examples of these meetings are as follows.

1. Pre-Start-Up Meeting

2. Post-Start-Up Meeting

3. Post-Installation Meeting

The final meeting that GE will schedule as part of the gas/steam turbine installation is the Post-Installation Meeting. The Post-Installation Meeting is a formal conclusion of all installation activities, and includes a presentation and review of installation data, tools, construction prints, etc. The Post-Installation Meeting is normally held one month after the commercial operating date.

B. Service Engineering Function

General Electric's Service Engineering function provides to the Purchaser all aspects of installation and maintenance throughout the life of the gas/steam turbine unit. This function is the focal point of all service related activities between GE and the Field Engineer.

General Electric is fully committed to the optimum operation and maintenance of each unit. Each gas/steam turbine user has an assigned Service Engineer responsible for fulfilling service functions. Throughout the life of the unit, the Service Engineer and other support personnel will work to provide recommendations on the latest operating and maintenance practices, meet with you prior to scheduled outages to assist in development of the outage scope and plan, and to make recommendations for replacement parts, improvements, and modifications. The Service Engineer can be contacted through the I&FS District Office.

VI. APPENDIX A

NOTICE OF LOSS OR DAMAGE

*To: _____ _____ _____ _____ _____ *(*Component responsible for filing Claim*)	ISSUED AT: _____ DATE: _____

Notice is hereby given to your component of loss or damage to the following articles from shipment received _____ _____ .

NO. AND TYPE OF PKGS. SHORT OR DAMAGED	DESCRIPTION OF ARTICLES SHORT OR DAMAGED	NATURE AND EXTENT OF LOSS OR DAMAGE

Consignee:		Destination:	
Consignor:		Shipping Point:	
Date of Shipment:		Route:	
Freight Bill No.:	Date:		Car/Trailer No.:
Car/Trailer Seals:			
Inspected by:		Carrier:	Date:
Inspection Report Attached:		Yes	No
Remarks:			
Req. No:	M/S No.:	R/M:	Lot No.:
		By:	

EXTRA WORK AUTHORIZATION

(Internal)

From: GE Energy Services

To: GE Product Service Dept.

Gentlemen:

This is a request for authorizing extra work charges at _____
for the workscope and cost estimate detailed below:

WORKSCOPE: _____

COST ESTIMATE:

Craft Labor _____ Hrs @ _____	$ _____	FSR #	_____
Equipment Rental:	$ _____		
Material:	$ _____	Field Engr Hours:	
Subtotal:	$ _____		_____
Markup:	$ _____		
TOTAL:	$ _____	Per Unit	

Please review the above request and provide comments.

Yours Truly,

Project Technical Director:	
Charge #:	Approval Signature: _____
FRN:	
Date:	
Comments:	

To: _____ **From:** _____

Pages to Follow _____

FRN No.: _____

RAPIFAX/FORM	
NOTES:	FSR No.
	CUST. UNIT #
	TB SER #
	FIRED STS HRS.
	DATE OPENED
	DATE CLOSED
	TLX REF #
	TYPE: (EMER.) (ROUTINE) (RUSH)
	GTD COMPLAINT #
	OTHER

DATA SECTION

ML ITEM # _____ ML DWG. # _____

REF. DWG. # _____ QTY. REQ'D _____

NOMENCLATURE _____

INFORMATION SECTION

_____

_____

_____

_____

_____

_____

_____

CC: _____ TECH ADV:. _____

_____

INSTALLATION RAPIFAX NO: (518) 385-4-81 (USA) or (518) 385-1677